



Using machine learning to help find paths through the map in Slay the Spire

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Abstract—Slay the Spire is a complex deck-building and roguelike game with many possibilities of improving players ability to win. An important part of Slay the Spire is choosing a path that makes the players character as successful as possible. In this study we show that machine learning can help players pick better paths by creating an Artificial Neural Network (ANN) that predicts the most successful path of all available paths, we also discuss what makes a path successful. This study performed two experiments, one user study and one simulation experiment, with the intention of evaluating the created ANN and analysing what makes paths successful. Through the user study this paper shows that the ANN was effective at predicting paths, outperforming all other human players who played normally in all three cases. This study concludes that machine learning can be used effectively to help make pathing decisions in Slay the Spire. Furthermore the study proves the importance of the room types 'Elite' and 'Campfires' through the simulation experiment, user study and analysis of data from previous playthroughs.

Index Terms—Artificial Intelligence; Artificial Neural Networks; Game; Machine learning; Pathfinding; Slay the Spire.

I. INTRODUCTION

Slay the spire is a popular card game, with roguelike [1] and deck-building [2] mechanics, that was released officially in 2019 after two years of early access. In Slay the Spire the player's goal is to survive through three procedurally generated levels (acts) and defeat the bosses at the end of each of them, if the player dies they have to start over from the beginning. An act consists of a procedurally generated map of connected rooms, with different functionality, with the last room being a boss room. Defeating or progressing through the different rooms adds to the players combat strength through different ways such as adding cards to their deck, healing the player or adding powerful permanent effects called relics. A path is a series of connected rooms that takes the player from the start of the act to the boss. It stands to reason that if different rooms add to the players combat strength in different ways then choosing a path that optimizes the players combat strength is important to the overall success of the playthrough. An example of how the map with connected rooms can look like is presented in figure 1.

RQ1: How can machine learning help predict successful paths in Slay the Spire?



Fig. 1: Example of the map with different rooms represented by different symbols and the connections between them in the form of lines

RQ2: What do successful paths have in common with each other?

Machine learning is an Artificial Intelligence (AI) system that, based on training data, makes a mathematical model that helps with decisions or predictions [3]. The training data is used to teach the algorithm to solve problems through different learning approaches such as supervised learning, unsupervised learning or reinforcement learning [3]. It is used in a variety of different fields such as games, chemical science and economics [4][5][6].

Machine learning has been used in different games to for example help build AI systems that play the game against professional human players. Open AI et al [7]. build an AI called OpenAI Five, whose architecture is a recurrent neural network that is trained with reinforcement learning, that plays Dota 2 and it managed to beat the best professional Dota 2 esports team. A famous example is Silver et al [8]. that built a deep neural network AI, AlphaGo, that plays the complicated game Go and it managed to beat a professional human player, which is something that was not thought to be possible for another decade. When it comes to specifically Slay the Spire, Alex Driedger made an artificial neural network to help with making reward choices after battles [9].

To answer the research questions this study follows a design research methodology [10] and performs two experiments. The

first experiment is a user study where participants play Slay the Spire either with help from an artifact developed to help with choosing a path or without help. The final results from the playthroughs with or without help are compared to see if the artifact is effective. The second experiment is a simulation experiment where the developed artifact predicts a path in multiple different playthroughs of the game, these paths can be compared to see if there is any commonality in how the artifact predicts.

The contribution of this paper is an artifact in the form of an Artificial Neural Network (ANN) that predicts a path for the player to take through the acts of Slay the Spire. The success of the artifact is evaluated by an experiment in the form of a user study where the results from playthroughs with the help of the artifact are compared to playthroughs without it. Furthermore this paper contributes to greater understanding of what is important when choosing paths in Slay the Spire.

II. RELATED WORK

A. Machine learning

Machine learning is a branch of AI focused on building applications that learn from data and improve their accuracy over time without being programmed to do so. There are three steps when developing a machine learning application. The first step is to select a training data set, which is data that is relevant for the solution of the problem. The second step is to choose a learning algorithm that will run the training data. There are many learning algorithms with a few examples being regression algorithms, decision trees and neural networks. The third step is to train the algorithm by running it over and over again while using the training data as input and comparing the output of the algorithm to what it should have produced based on the training data. The result of the third step is a machine learning model that can predict or decide based on inputs provided [11].

When it comes to games machine learning has been used in a variety of different ways. Siang et al [12]. made a machine learning application that can learn game-playing strategies of the game Othello. They utilized a evolving neural network as their machine learning approach and they were successful in making it learn how to play Othello.

Geisler [13] explores how machine learning algorithms can help with behavior modeling in first person shooter games. They first collect data by having an expert player play through a modified version of "Soldier of Fortune 2". Then they developed and trained an ANN with the data collected and they were successful in developing an agent that had learnt combat behaviour from the expert player.

B. Artificial Neural Networks

An Artificial Neuron Network (ANN) is a computational model based on the structure and functions of biological Neural Networks. An ANN is made of multiple layers of connected neurons (nodes) that start with the input layer and ends with the output layer. In between the input layer and the output layer, one or more hidden layers are placed. A neuron

receives inputs from the connected neurons of the previous layer or from a data set if the neuron is in the input layer. Every connection between neurons has a weight that can be adjusted by the algorithm when training. When the neuron has received all inputs it processes by multiplying each input with the weight of the connection and running it through an activation function. Either the activation function activates and the output from the neuron is one or it does not activate and the output is zero. That output is sent through the connections to neurons of the next layer [14][15].

When training an ANN, training sets of known inputs and result are applied to the algorithm iteratively. With each iterative pass through the algorithm the difference between the output of the algorithm and the desired result found in the training set is calculated, this is the error of the algorithm. The algorithm then adjusts the weights of its connections based on a learning rule and the error value of the algorithm [14][15].

1) *Multilayer perceptron:* The Multilayer Perceptron (MLP) is a type of ANN that consists of at least three layers: the input layer, one or more hidden layers and the output layer. Every neuron that is not in the input layer has a nonlinear activation function that it uses when calculating its output to the next layer. The MLP is fully connected which means every neuron of a layer is connected to every neuron of the next layer. Training for a MLP is carried out through the learning rule backpropagation [16].

C. Previous research on Artificial Intelligence that plays games

1) *Dota 2 with Large Scale Deep Reinforcement Learning:* The study by Open AI et al [7]. shows how reinforcement learning can be used to train an AI that beats the best professional esports players of the game Dota 2 and Open AI et al. hypothesize in their conclusions that the same techniques can be used in other esports. Dota 2 is an interesting game for this type of research since it has complex rules, a lot of game logic that the developers have worked on for years, long time horizons, partial observability, and high dimensionality of observation and action space. It is a game with a large professional esports scene which provided good players for the AI to learn from.

In simple terms the AI, called OpenAI Five, acts every fourth frame which the developers call a timestep. In each of these timesteps OpenAI Five observes the game states different variables, for example their health and enemy positions, and processes them through the already trained neural network to then return a discrete action to the game engine in the form of a movement, attack or any other action available. The network is a recurrent neural network that has approximately 159 million parameters and it primarily consists of a single-layer 4096-unit LSTM (Long short-term memory) architecture. The network is trained by repeatedly playing the game and passing the current observation variables as inputs to the network and sampling an action from the output distribution at each timestep.

The goal Open AI et al. has for their study is to find a policy that maximizes the chances of defeating professional

human experts of the game. They do this by maximizing a reward function, this includes signals such as characters dying, collection resources, etc. The reward function is used when training the network to send positive signals to the network when it makes a good decision and negative signals when it makes bad decisions. Open AI et al. creates the reward function at the beginning of the project based on the experiences in the game of the project members and that reward function works fairly well and only needs a few tweaks during the project.

After OpenAI Five is finished training it plays an exhibition match against the best professional Dota 2 team and defeats them 2-0. Furthermore Open AI et al. challenge the Dota 2 community to see if they can beat OpenAI Five, the result is that OpenAI Five wins 99,4% of the 7257 games it plays against the community. The conclusions Open AI et al. makes is that reinforcement learning can make an AI that achieves superhuman performance in competitive esports games and they hypothesize that the same approach they used can be used in other esports games with similar results.

2) *Mastering the game of Go with deep neural networks and tree search:* The study by Silver et al [8]. shows how a deep neural network and tree search artificial intelligence is capable of mastering the game of Go. Go is a very challenging game for artificial intelligence because it has an enormous search space and it is very difficult to evaluate board positions and moves. It is a game with perfect information where there is no hidden information available, all variables are visible the entire game. This means there is an optimal value function that can predict the optimal move at all times based on the information, this study by Silver et al. tries to find that function but it is hard due to the complexity of Go and its variables.

Previous approaches to playing Go with artificial intelligence uses Monte Carlo tree search (MCTS) algorithms to find the next move. The study by Silver et al. combines MCTS with deep neural networks to achieve better results. The neural network is trained in separate steps with different learning approaches. Firstly a policy network is trained by supervised learning where information for learning is gained from expert human players moves. This policy network is improved by a reinforcement learning policy network that is trained from games of self play. The final network is a value network that predicts the winner between the reinforcement learning network and itself. AlphaGo is the final AI program and combines these networks with a MCTS to choose which moves to make when playing Go. The networks help reduce the effective depth and breadth of the search tree. The value network helps with evaluating positions on the board and the policy network helps with sampling actions.

Silver et al. evaluated AlphaGO against previously made AI systems that play Go and it won 494 out of 495 games against these other AI:s. AlphaGo then defeated a professional player 5-0 which is the first time an AI has done this in the game of Go, a feat previously thought to be a decade away.

3) *AlphaStar: Mastering the Real-Time Strategy Game StarCraft II:* The study by The AlphaStar team [17] shows how

a deep neural network AI can defeat professional players in the game StarCraft II. StarCraft II is considered one of the most challenging real-time strategy games and an interesting challenge for AI developers. The game is very complex and before The AlphaStar team no one had successfully defeated a professional player with an AI. In StarCraft II the player needs to balance short and long-term goals and adapt to unexpected situations which is hard for AI systems that usually are brittle and inflexible. Add to that some information is hidden in StarCraft II, it is also a real-time game, that means there are no pauses for the AI to think and StarCraft II also has a large action space due to it being in real-time with the player forced to control multiple units simultaneously.

The network built by The AlphaStar team uses raw game interface data (for example list of units and their properties) as input data for their AI AlphaStar, the output from the network is a sequence of instructions that constitutes an action within the game. The network structure of AlphaStar applies a transformer torso to the units, combined with a deep LSTM (long short-term memory) core, an auto-regressive policy head with a pointer network and a centralised value base line. All this is first trained with supervised learning from data of previous games of StarCraft II released by the company that makes StarCraft II: Blizzard. This trained network is then used in an interesting way to perform reinforcement learning. The AlphaStar team built a league of competing agents (versions of the trained network), then these agents competed against each other in this league. This means they learn similar to human players on a ranked ladder, the agents develop strategies and then counter strategies to those strategies. To create diversity the agents have different learning objectives, this could be for example beating a specific strategy, that are adapted during the training process. The league ran for 14 days and every agent played approximately 200 years of real time StarCraft II during the league. The final agent is made with the most effective mixture of strategies that have been discovered by the agents of the league, it also has to run on a single GPU and nothing heavier.

The AlphaStar team evaluated the AI AlphaStar against two professional players in a regular match. Against both players AlphaStar won 5-0 which is the first time an AI defeated a professional player in StarCraft II.

4) *Machine learning in Slay the Spire:* There is not much research done about Slay the Spire but there is one interesting research piece done by Alex Driedger [9], where he utilizes ANN to predict how much damage a player should take each fight. He first made his own training sets from data provided by the website "spirelog.com" where players themselves can register playthroughs of the game. Then he made an ANN that can predict how much damage the player should take each fight based on the player's cards, relics and what enemy they are fighting. He also used this model to evaluate cards, when choosing to add them to the players deck, by comparing the result of a fight without the card in the deck and a fight with the card in the deck.

III. TECHNOLOGY USED

A. TensorFlow

TensorFlow is a machine learning system that is useful when developing ANN:s since it already has clear data structures and developed methods for learning and training ANN:s. It is an open-source project developed by Google Brain [18].

When using TensorFlow the first step is to build an input pipeline for the project to convert the training data to data structures that TensorFlow uses and this is done with the help of the "tf.data" library [19]. Usually it is the data structure "Tensor" which is used [20]. Tensor is a data structure that is a multidimensional array of elements and is used in many of TensorFlow's methods.

After the input pipeline is built and the training data is stored in the correct data structures it is time to train and evaluate the model. The API used for training and evaluating models is the "Keras" functional API. With "Keras" the user can choose how many layers the network should have, what type they should be and what activation function to use with each layer. The built in training loop for "Keras" is called "fit" and the user can customize that loop to suit how they want the training to be. The model that is created is evaluated by the method "evaluate" with specified test data and the user can specify how many epochs they want to use [21].

TensorFlow allows for a lot of customization in each step and it has a big library with description of methods and tutorials with varying degrees of complexity [22][23].

IV. SLAY THE SPIRE

A. Terminology

- **Relic** Objects found in the game or started with that provides the player with powerful passive abilities that can help in different ways. Another large factor for winning the game is having good relics and potions. An example is the relic that gives the player more energy to use in a fight which increases the player's chances to survive the fight and continue on to win. Another example is the "wing boots" that makes the player able to jump to different nodes not directly connected to the one the player is staying on. This is a more indirect effect since it doesn't give the player any edge in combat but instead gives the player a greater control in choosing what nodes to visit to get ones with the highest value.
- **Energy** The resource cards use when played in combat, if no energy is left then the player can't play any cards requiring energy. The player starts with three energy each round of combat and can't get anymore unless a relic that can provide more energy has been picked up.
- **Gold** A resource used to pay for services from the shop in the game.
- **Potions** Items that are used in combat providing positive effects for the player or negative effects towards targets when used depending on the potions used. Is consumed (discarded) when used and because of this, not very effective for the whole game but very useful when

used at important key points in fights with larger threats like a boss or elite enemy.

Slay the spire is a card based roguelike video game. The objective of the game is to reach the last level and win over the end boss there. To reach the boss the player traverses through a map composed of nodes connected to each other. Each node visited triggers an encounter of some kind which can present the player with a challenge or resource for becoming stronger. The player has to fight in some of these nodes to continue by using the deck of cards each class starts with and the resources they have gathered while visiting nodes. The resources the player can gather are the following: relics, potions, cards and gold. The player gets to choose one of four different classes to start the game as. Each class provides the player with a unique set of cards and relics only that class uses. After winning the game for the first time the player can choose something called ascension level which increases the game's difficulty for example by making the enemies stronger or the player starting with less HP.

B. The map

To reach the end boss the player has to move through a map (see figure 2) consisting of different nodes connected to each other in different configurations. The player can only walk onto nodes in front of them that are directly connected to the node the player is currently staying on. Each node contains one of seven different rooms that can be encountered: enemy, campfire, elite, shop, random event, treasure chest or boss.

- Enemy rooms contain a basic enemy a normal player can dispatch fairly easily with a new card and a small amount of money as reward.
- Campfire rooms are a room that give the player two options when they enter, either the player restores 15 percent of their maximum health or they choose one card from their deck to upgrade.
- Elite are enemies that are much more dangerous but rewards the player with a card of higher rarity than a normal enemy would, a medium amount of money and a relic which gives the player a beneficial passive effect.
- Shop is a room containing a merchant, where the player can spend the gold gathered from different encounters to buy certain cards, artifacts, potions or remove a card from their deck. The available cards or items to buy are randomly chosen for each shop.
- Random events are a group of random encounters where the player is given a choice between different options they can choose from resulting in different outcomes depending on what the player chooses. The outcomes can be good, bad or both at the same time depending on what choices the player makes.
- Treasure chests are rooms containing a chest of varying quality where the player receives a relic upon entering the room. The quality of the chest gives the relic inside the same quality, a higher quality chest will contain a more powerful relic.



Fig. 2: Example of the starting area of a map. The player has many different choices to choose from. See the Legend in the image for the room symbols

- The boss room is an enemy room containing the boss itself that the player has to defeat to gain access to the next level and are rewarded when defeating the boss with a very powerful relic, a rare cards and a large amount of gold.

The player walks through the nodes with different encounters until they reach a boss for the current level they are in. If the player wins over the boss then they move on to the next level. The player wins the whole game by defeating the boss at level three.

C. Cards

One of the major obstacles of the game is rooms containing enemies. Enemies can reduce the player's health points (HP) which if it reaches zero means game over. Game over means the player will have to start all over again with a new run, losing all progress. To combat this, the player fights the enemies in turn-based combat where the player can play cards (see figure 3 for a example) with various effects to destroy the enemies and minimize HP loss. Cards are drawn each round and discarded when the round is over. Each card has a cost to play it in another form of resource called energy. The player can find more and better cards as rewards for various things while playing. Collecting these cards and putting together a deck with cards that are powerful and have a good synergy together is one of the primary ways becoming stronger to reach further and win the whole game.



Fig. 3: Example of the card Strike, the number in the top left corner is the associated energy cost with playing the card.

D. Classes

Every time a new game is started, the player has to choose one of four different classes to play as. Each class gives access to a set of unique cards and items only available to that class. The strategies in combat for each class becomes very different due to this and changes what cards and items the player searches for to become stronger. Written here is a small summary of each class and their strategies.

- Ironclad: Focusing on health, raw damage and using cards once and then throwing them away.
- Silent: picking up cards, using deliberating effects on enemies and setting up attacks with many different cards

- Defect: Has a special game mechanics called orbs. The player is surrounded with orbs that trigger an effect every round. The player can create more orbs from cards or use them up in one burst. Requires some planning and reading the flow of the fight to plan for how to use them.
- Watcher: Uses a special mechanic called stances that they change between by playing cards. The stances are calm and rage, calm gives the player more energy and rage makes the player deals twice as much damage but receive twice as much damage too. This class is a high risk high reward kind of class in which fights can end in two rounds if played well or the player could lose the whole game.

E. Ascension

After winning the game the first time the player unlocks a new difficulty for the game called ascension. Ascension is a level number the player chooses when starting a new game and makes the game harder in different ways as the level increases. At level 1 elite enemies appear more often, in level 2 normal enemies are more deadlier and so on. The levels difficulty modifier is cumulative so when the game start for example at level 2, it will both have more elite enemies and contain deadlier normal enemies.

V. RESEARCH METHODOLOGY

The methodology used for the proposed research question was *design science research* [10]. Design science research is a model consisting of guidelines for how research should be done. Following these guidelines helps researchers for example do research that is reproducible, makes it more legitimized and helps them accept research that is well done using understood and accepted processes.

A. Problem identification and motivation

To play Slay the Spire and win the game, a path has to be walked by the player from a map of nodes. A path is the nodes you choose to travel through in the graph to reach the end node. Each node in the graph is some kind of encounter the player can interact with. Depending on what they land on, the player can encounter an enemy, elite, shop, treasure chest or special event. Choosing this path is an balance act for the player since they have to keep track of resources like how much HP they have left or if they think they need more cards to their deck. An example of this balance act is trying to encounter as many elites as possible. Since elites drop items that can increase the chances of winning this means the player will have an easier time in the later parts of the game but may also lead to game over if encountered while the player is low on health. On the other hand, normal enemies are a safer options but doesn't give as good items to help the player win the game. This balance act of different variables is a difficult problem to solve since the player has to plan the path to choose that will give them the highest chance of winning but doesn't want to make it too difficult and loose after only a few encounters. On the other hand, if the player chooses to encounter as many difficult encounters as possible with high

rewards and manages to survive, then their chances of winning the whole game will be a lot higher.

B. Define the objectives for a solution

The developers of Slay the Spire has recorded copious amount of data open to the public which makes it possible to train an Artificial Neural Network (ANN). The goal is thus to create an artifact that can assist humans with picking alternative paths. Assisting humans picking a path means picking paths that has the player make it further or the same distance than they would have done without the artifact. The developed artifact should be able to format the data for use in training the neural network, train a neural network to pick path in Slay the Spire and be able to connect the neural network with the game itself to read the current games data to come up with new paths for the player.

C. Design and development

The artifact is developed using python, java and the framework TensorFlow (see the section technology used for more info) and is composed of a program formatting the data for training, a mod to Slay the Spire connecting the game to the ANN providing it with input data and lastly the ANN itself that will pick what paths to use. The formatting program is written in C and is needed since data needed for training the ANN is clumped together in large files containing all other information recorded by the developers. Removing unnecessary data reduces the time it takes for the training, file sizes and readability. There is also a considerable amount of data that is from players cheating that has to be removed. If this data where to be use for the training then the ANN might be trained the wrong way to make decisions that aren't possible. The program providing the ANN with input data is written in java because the modding framework connected to Slay the Spire that can access the game data is written in java. This program continuously provide the ANN with input data for the current game with variables like player HP, amount of gold, path traveled, current deck and current relics. The artifact needs this so the ANN can make decisions based on the data it receives. The ANN itself is written in python and is constructed from the framework TensorFlow that gives access to all the different functionalities a ANN requires without having to program one from scratch.

D. Demonstration

The ANN was trained with training data provided by the developers and process by us to only contain the data needed for training the output after which the artifact was used by a player playing the game, to fight the best they can. As it is time to decide what node they shall travel to in the map, instead of choosing themselves, the artifact pick the path it thinks will give the player the highest chance of winning. The player continues playing the game like normal until it's time to pick a path again in which the artifact picks the path again. This cycle continues until the player lose or win the game.

E. Evaluation

The trained neural network is evaluated using A/B testing which is a method where an experiment is performed in two variants to test a subjects response to the different variants [24]. This method is used because it's very fitting to use when you want to compare if the developed Artifact is better at providing a path than a human. The test is done in two groups, one control-group consisting of only humans and another group of humans assisted by the ANN. All the groups play the game with the character ironclad and try to get as far as possible without dying. The reason for only playing with the ironclad character is to minimize variables that can affect the experiment. Since a playthrough of Slay the Spire can take quite a long time the participants only play one time each, even though the drawback is that the success or failure of their playthrough could have been impacted by luck and random elements. Another version of the experiment would be to have the same participant play the game two times on the same seed, first without help from the artifact and then with help from the artifact. However that experiment version is not chosen because in Slay the Spire the decision making of the player changes heavily if they know anything about the seed of the playthrough. For example what elites they will fight and what cards they will be shown as rewards.

The ANN group will decide the path they take by choosing what the ANN tells the player to choose. The control-group is used to evaluate if the ANN contribute in any way to the rate of winning, this indirectly tells if the ANN is better at finding a path than a human. The person being tested is supervised by a researcher to help in case an unexpected problem arise or the tester needs directions. The researcher asks questions during the test about the testers decisions while playing and record them. The protocol with the questions the researcher asks is in Appendix C. After the tester finishes a run of the game if they where part of the group which uses the ANN and not all of the questions where asked and answered during the session they are asked to fill in a questionnaire, visible in Appendix B, with questions about their opinions regarding the ANN and what decisions they would have done if they where instead allowed to choose the path. The researcher responsible for the test notes down the seed for the map, how far the tester reached, the path they took, how they died and the answers to the questions in the protocol for a session. This data is recorded for both groups of testers.

To find answers for the second research question what makes a good path, a small quantitative study on the output of the ANN is also performed. For this second test the ANN is activated 10 times in different games with different seeds without playing though the games, instead just the data from the output in the beginning of the first floor is gathered. The calculated path is recorded by the researcher who furthermore also records if the chosen path contained the most rooms of elite or not of all possible path provided to the network.

F. Communication

The artifact tries to solve an interesting problem regarding how an ANN can help predict a path in slay the spire. Analysing the ANNs pathing decisions can lead to new insights regarding game design and balance between different factors in games. The artifact could be of further practical use as a automatic tester for testing paths in games using graphs.

VI. ARTIFACT

The artifact we are developing is an ANN that predicts a path through the map in Slay the Spire based on provided inputs. It is written in the programming language Python and utilizes the machine learning system TensorFlow. TensorFlow has many methods that help with making complex ANN:s.

The design of our ANN is a Multilayer Perceptron (MLP) and is described further in the related work section. Our MLP consist of three layers: the input layer, a hidden layer, the output layer and before the input layer something that is not quite layer but which is done for all inputs we call the preprocessing layer. The input layer consist of 24 input nodes, Although a large amount of inputs, 15 of the nodes is just for one datatype the path chosen. The reason for this is because the value for path taken was difficult for the preprocessing layer to transform into fair and usable data the network could make decisions based on. The networks performance greatly improved when divided like this.

The prepossessing layer is needed to prepare the input data to the right format since it is sent from the game in many different forms that can't be read by the network which works by passing on scalar values from one layer to another. The preprocessing layer works by first transforming all inputs of the string type into a scalar value and then normalize all values that now are scalar values. For example we have an input for a "deck" value that are a string who first gets converted to a number with algorithms developed by the programmers who also created the tensor flow library. This number is then normalized with all other deck input so its value is between 0 and 1. Now it is ready to be used by the network either during the game or while training. All this preprocessing is done for every value the network uses.

The activation function we are using is the ReLU activation function which is the most widely used activation function. The ReLU function is defined as: $f(x) = \max(x, 0)$ [25]. What this means is that, if the input is positive the ReLU function outputs it directly, otherwise it outputs a zero. Our network is trained by using the supervised learning.

The inputs for our network are the following: "character chosen", "ascension level", "floor level", "path taken", "hp per floor", "gold per floor", "deck" and "relics". Under here is an list of all the inputs with a motivation why they are relevant to use for in our network and their value type:

- "Character chosen" is used to evaluate when choosing paths since the classes play differently and need different things from their path. The form if this of this input is a number between 0 to 4 representation what character the player choose to use.

"Ascension level" for the input is relevant because it is the best variable for determining the skill level of the player and is a value in the form of a number ranging from 0 to 20 where 0 is playing without any ascension on and 20 is at the highest difficulty with all negative obstacles activated.

- "Floor level" might be relevant to the input since each floor contains different events and enemies getting progressively harder which can have a large effect on the pathfinding. The floor level is a number ranging between 0 to 4 where 0 is the start level and 4 is the last level containing the last boss.
- "Gold per floor" and "hp per floor" are relevant to pathfinding since they might have a big impact on the networks decisions since the player might alter their path if they are low on hp or have a lot of gold with a shop nearby. Both values are number values where gold per floor can vary between 0 to 600 and "hp per floor" varies between 1 to 150.
- "Path taken" is the most important input since the whole purpose of the network is to pick a path. Its value is a string containing the path currently taken and look like this: "M-?-?-?-R-?-E-T-?-M-M-?-?-R-BOSS".
- The "Deck" input the player uses has a huge impact on how players survives combat. Without a good deck the player will lose immediately so knowing how well the player would fare against strong enemies in the next rooms is crucial for the network. This input is also a string due to the deck being in a constant state of change as the player is continuously defeating enemies which rewards them with new cards or some other room that have an effect on the deck.
- Last input is "relics" that fills many of the same criteria the "deck" input does but also has some different things about it. It affects survivability in combat like decks but can also affect things like gold which doesn't directly affects combat but lead to the player becoming stronger. Taking the right paths when carrying a specific relic can lead to a new strategy needed for choosing path due to the power the relics gives the player. For example there is a relic that gives the player a lot of extra gold each time they visit a room that isn't a shop. When they visit a shop instead the relics is destroyed. Getting a relic like that can change the most effective path drastically since picking a path with a shop might lead to a lot less gold which is why the relics is needed as an input.

All the inputs are provided to the ANN at the start of each act when a new map is provided. Then the inputs are queried again after every room to see if the path is changed based on the new inputs (shown in Figure 3). The mod finds all possible paths through the map and prints them together with the other input variables to a csv file, these paths are then one by one utilized as inputs for the ANN. The path that receives the highest prediction chance of victory becomes the recommended path. The result is displayed graphically in the

ANN program.

The output layer contains only one node that is a binary variable "victory", which can either be "true" or "false" by changing between 1 and 0. The ANN predicts how likely the result is either "true" or "false", for example the result can be 70% "true" and 30% "false" with the output value of 0,7. We had considered to use the amount of rooms traveled as an alternative output since it would give a more accurate indicator on how well the current evaluated path would fare but it was decided to not use this because we did not manage to make it to work in the time we had left.

The network is trained with 172203 rows of data that the developers of the game had gathered from real players. The row of data in the data file looked like the following "character,ascension,floor,hp,gold,room1,room2,room3,room4,room5,room6,room7,room8,room9,room10,room11,room12,room13,room14,room15,room16,room17,room18,room19,room20,room21,room22,room23,room24,room25,room26,room27,room28,room29,room30,room31,room32,room33,room34,room35,room36,room37,room38,room39,room40,room41,room42,room43,room44,room45,room46,room47,room48,room49,room50,room51,room52,room53,room54,room55,room56,room57,room58,room59,room60,room61,room62,room63,room64,room65,room66,room67,room68,room69,room70,room71,room72,room73,room74,room75,room76,room77,room78,room79,room80,room81,room82,room83,room84,room85,room86,room87,room88,room89,room90,room91,room92,room93,room94,room95,room96,room97,room98,room99,room100,room101,room102,room103,room104,room105,room106,room107,room108,room109,room110,room111,room112,room113,room114,room115,room116,room117,room118,room119,room120,room121,room122,room123,room124,room125,room126,room127,room128,room129,room130,room131,room132,room133,room134,room135,room136,room137,room138,room139,room140,room141,room142,room143,room144,room145,room146,room147,room148,room149,room150,room151,room152,room153,room154,room155,room156,room157,room158,room159,room160,room161,room162,room163,room164,room165,room166,room167,room168,room169,room170,room171,room172,room173,room174,room175,room176,room177,room178,room179,room180,room181,room182,room183,room184,room185,room186,room187,room188,room189,room190,room191,room192,room193,room194,room195,room196,room197,room198,room199,room200,room201,room202,room203,room204,room205,room206,room207,room208,room209,room210,room211,room212,room213,room214,room215,room216,room217,room218,room219,room220,room221,room222,room223,room224,room225,room226,room227,room228,room229,room230,room231,room232,room233,room234,room235,room236,room237,room238,room239,room240,room241,room242,room243,room244,room245,room246,room247,room248,room249,room250,room251,room252,room253,room254,room255,room256,room257,room258,room259,room260,room261,room262,room263,room264,room265,room266,room267,room268,room269,room270,room271,room272,room273,room274,room275,room276,room277,room278,room279,room280,room281,room282,room283,room284,room285,room286,room287,room288,room289,room290,room291,room292,room293,room294,room295,room296,room297,room298,room299,room300,room301,room302,room303,room304,room305,room306,room307,room308,room309,room310,room311,room312,room313,room314,room315,room316,room317,room318,room319,room320,room321,room322,room323,room324,room325,room326,room327,room328,room329,room330,room331,room332,room333,room334,room335,room336,room337,room338,room339,room340,room341,room342,room343,room344,room345,room346,room347,room348,room349,room350,room351,room352,room353,room354,room355,room356,room357,room358,room359,room360,room361,room362,room363,room364,room365,room366,room367,room368,room369,room370,room371,room372,room373,room374,room375,room376,room377,room378,room379,room380,room381,room382,room383,room384,room385,room386,room387,room388,room389,room390,room391,room392,room393,room394,room395,room396,room397,room398,room399,room400,room401,room402,room403,room404,room405,room406,room407,room408,room409,room410,room411,room412,room413,room414,room415,room416,room417,room418,room419,room420,room421,room422,room423,room424,room425,room426,room427,room428,room429,room430,room431,room432,room433,room434,room435,room436,room437,room438,room439,room440,room441,room442,room443,room444,room445,room446,room447,room448,room449,room450,room451,room452,room453,room454,room455,room456,room457,room458,room459,room460,room461,room462,room463,room464,room465,room466,room467,room468,room469,room470,room471,room472,room473,room474,room475,room476,room477,room478,room479,room480,room481,room482,room483,room484,room485,room486,room487,room488,room489,room490,room491,room492,room493,room494,room495,room496,room497,room498,room499,room500,room501,room502,room503,room504,room505,room506,room507,room508,room509,room510,room511,room512,room513,room514,room515,room516,room517,room518,room519,room520,room521,room522,room523,room524,room525,room526,room527,room528,room529,room530,room531,room532,room533,room534,room535,room536,room537,room538,room539,room540,room541,room542,room543,room544,room545,room546,room547,room548,room549,room550,room551,room552,room553,room554,room555,room556,room557,room558,room559,room560,room561,room562,room563,room564,room565,room566,room567,room568,room569,room570,room571,room572,room573,room574,room575,room576,room577,room578,room579,room580,room581,room582,room583,room584,room585,room586,room587,room588,room589,room590,room591,room592,room593,room594,room595,room596,room597,room598,room599,room600,room601,room602,room603,room604,room605,room606,room607,room608,room609,room610,room611,room612,room613,room614,room615,room616,room617,room618,room619,room620,room621,room622,room623,room624,room625,room626,room627,room628,room629,room630,room631,room632,room633,room634,r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A. Slay the Spire mod

Slay the Spire has a big modding community and they provide easily accessible tools to develop your own mod. We base our mod on the Default Mod Base made by Gremious which is found on github [26]. It is an over-commented mod to Slay the Spire that acts as a tutorial helping us understand how we can set everything up and what we can change to make it behave as we want. First there are two mods to Slay the Spire that need to be downloaded in order to make any subsequent mod work, these are Mod the Spire [27] and the Base mod [28]. They provide the ability to easily connect to the source code without having to understand or read it by making easily understood methods that act as bridges between the source code and the mod being made. They create what is referred to as hooks to the source code and the mod being made can subscribe to these different hooks. For example if the programmer of a new mod wants something to happen after a battle is finished they subscribe to the hook `PostBattleSubscriber` and implement the `RecievePostBattle` method. In the `RecievePostBattle` method they write what they want to happen after a battle is finished.

In our mod we print the variables we are interested in from the game, as different columns, to a csv file that is later read by the ANN program. These variables are player class, ascension level, current health, current gold, deck, relics and each room on the path towards the boss as individual columns. All variables except the path are easily accessible by calling on the abstract variables `AbstractDungeon`, which stores information about the dungeon, and `AbstractPlayer`, which stores information about the player. The path is harder since there are multiple possible paths to the boss and we want the ANN program to evaluate everyone of them. We want to get every possible path printed to the csv file therefore we print each unique path as its own row, where the other column variables are the same in all rows since they are the same for each path. The map of Slay the Spire is represented as a connected graph which means we do not have to use a complex search algorithm to make one, instead we use the methods and variables already in the program to find all possible paths from the current node (the floor the player is on) to the boss by traveling through each node's list of connected edges. The edges in a node's connected edges list are all forward in the map and not backwards towards previous nodes, this makes it easy for us to travel from node to node when finding possible paths and always know that we are moving towards the boss.

When the dungeon is initialized at the start of the act all possible paths from all starting nodes to the boss are collected and stored in a list, by traveling from node to node by the connected edges until the boss is reached, then a new path where different edges and nodes are visited is collected and stored. When a new node is traveled to (the player goes to the next floor) all paths in the list that do not contain the new node are removed from the list, this repeats every time the player travels to a new floor.

The hook we subscribe to when we want to print the

variables is `PostDungeonUpdate` that updates every time the dungeon changes which it does every time something in the dungeon is changed. We tried with `PostBattle` but the problem is that the receive method is called before the gold and reward for the battle has been collected and this meant the gold value that is printed to the csv file in that case was wrong. To not make the mod print every time the `PostDungeonUpdate` is called we have a few if statements that check if the variables have been updated which would mean we want to print.

VII. EXPERIMENT

The main experiment this paper conducts is a form of A/B testing [24] with participants divided into pairs, where one member of the pair plays through Slay the Spire while choosing the path themselves and the other member plays while choosing the recommended path from the ANN. There are three variables that are static for both members of the pair: the seed of the game, the starting class and the ascension level of the playthrough. By keeping these static the playthroughs of the participants are more similar to each other and the path they choose has more impact on the final result.

The participants in the experiment are chosen with varying degrees of experience and skill in the game. This is determined by how much they have played the game and what ascension level they have reached. It is impossible to match the skill level of the participants exactly since two participants that have reached the same ascension level and played approximately as much of the game may still have a skill difference but these two variables are the best indication of skill we have available. The participants do not need to own the game but they need to have played the game before and have enough understanding of the game to navigate combat and reward choices by themselves. When dividing the participants into different pairs we are ensuring that the members of the pair are evenly matched in terms of experience and skill. To better accomplish this we ask the participants to fill out a Google Form with basic questions about their experience in Slay the Spire before the experiment is carried out, these questions can be found in Appendix A. When contacting interested participants we ask for their permission to use the data gathered in our thesis and explain that all participants remain anonymous. The experiment is performed with six participants divided into three pairs.

The objective of the experiment is to compare how far the members of the pair reached in Slay the Spire during their experiment session. The intention is to find out if our artifact is effective and if the member using the artifact has a better result than the member choosing by themselves. During the experiment we ask qualitative questions to the participants with the objective of finding out what they think about when choosing paths.

The procedure of the experiment starts with gathering participants. This is done through asking students of Malmö University if they are interested, primarily students of the game design program are asked. Before the first session of the experiment we divide the participants into pairs where we try

to have as equal skill level as possible between the members of the pair and we randomly choose one member of the pair to be the one using the artifact as help. A session of the experiment is carried out with one participant and one observer, who is a contributor to this paper, remotely with the video call program Zoom using their built in remote control feature to provide the participant with remote control of the observer's computer mouse. This means only the observer needs to have the game, mod to the game and ANN program running on their computer since the participant can control the observer's mouse and play the game on the observer's computer. In the session the participant plays through the game like normal, deciding what cards to play and what cards to pick, only when choosing a path they either have help from the artifact or chooses by themselves. During the session the observer takes notes of events that happen and asks qualitative questions to the participant about what they think about when choosing a path, the protocol for a session with the questions that are asked is found in Appendix C.

A. Simulation Experiment

A second smaller experiment is done to seek answers to our second research question to find what connection successful paths had with each other. By analysing the paths the ANN think has the highest chance of winning we hope to draw a conclusions for what makes a good path to win the game.

The test is performed by the researchers since there is no need for testers due to the only recording data not being affected by human inputs.

The experiment is conducted by recording the suggested path the ANN gives for multiple runs of the game. It starts with feeding the ANN input data from the start of a normal run of the game. Once the suggested path is calculated by the ANN, the path is recorded and the current game is abandoned with a new game taking its place with a new seed. The new game is started, the best path the ANN computes is recorded and this cycle continues until 10 paths has been recorded from 10 different runs of the game.

VIII. RESULTS

The experiment was conducted in sessions with one participant at a time being observed by one of the authors of this thesis. The session was done online where the observer and participant communicated with the help of a communication program of their choice such as Zoom or Discord. The observer had the correct programs running on their computer and then they let the participant control their computer by a remote control program. The participant could then play through a run of Slay the Spire with the correct setup without having to install any programs themselves. The participants all played Ironclad on ascension level 0.

If the player was chosen to play without help from the network they played as normal. If instead they were chosen to play with help from the network they had to press the button on the User Interface (UI) of the program running the network after every floor when it was time to move on. After the new

path was calculated it was displayed on the same UI in text form and the participant had to try to match the path in text form to the correct path on the map in Slay the Spire (see figure 6). If the output from the network was M-M-?-E-R-BOSS then the participant had to find the path in Slay the Spire with those exact rooms in that exact order and follow it.

During the experiment the observer asked questions to the participant about what they thought about when they are choosing paths in Slay the Spire, these questions can be found in Appendix C. The observer also asked questions about the participants thoughts on the path provided by the network. The final result of the run, where the participant died and what path they took, was collected after the experiment from the run history section of Slay the Spire where information about previous runs are displayed.

A. Participant 1A

Participant 1A played with help from the AI and reached the last floor and won the whole game by beating the boss "the awakened one". The participant played on the seed: XWZX4NYBCMC8.

The different rooms of the path are represented by different characters. 'M' is monster room, 'R' is a campfire room, '?' is an event room, 'E' is an elite room, 'T' is a treasure room, '\$' is a shop room and 'BOSS' is the boss room of that act. The path the participant took:

Act 1: M-M-M-M-M-R-E-R-T-R-E-R-E-M-R-BOSS

Act 2: M-?-M-M-?-R-M-?-T-?-E-\$-M-E-R-BOSS

Act 3: M-M-?-M-M-E-M-E-T-E-?-R-M-M-R-BOSS

The participant had played the game for 67h, had reached ascension 3 and was very comfortable with the mechanics. The participant had a good tactic when choosing cards which led to them creating a strong deck with a high synergy between almost all cards. The participant thought the AI was too reckless when choosing paths since it never choose a new path to avoid strong enemies even when the player started to have very low health. All other choices the AI did the participant was satisfied with. If the participant had control over what path to take, then they thought they would have chosen to avoid some of the elite rooms while low on health instead of going to them like the AI did.

B. Participant 1B

Participant 1B played without help and reached the last boss fight on floor 50. They died in the last boss fight to the boss called "Awakened one". The participant played on the seed: XWZX4NYBCMC8. The path the participant took through the acts was the following:

Act 1: M-?-?-?-R-?-E-T-?-M-M-?-?-R-BOSS

Act 2: M-M-\$-?-?-R-?-R-T-M-E-M-?-E-R-BOSS

Act 3: M-?-M-?-R-M-E-T-E-?-?-R-M-R-BOSS

The participant had played Slay the Spire for 40 hours and had reached ascension level 3. They considered themselves fairly knowledgeable about the game and was comfortable with the mechanics of the game. The observer of the session

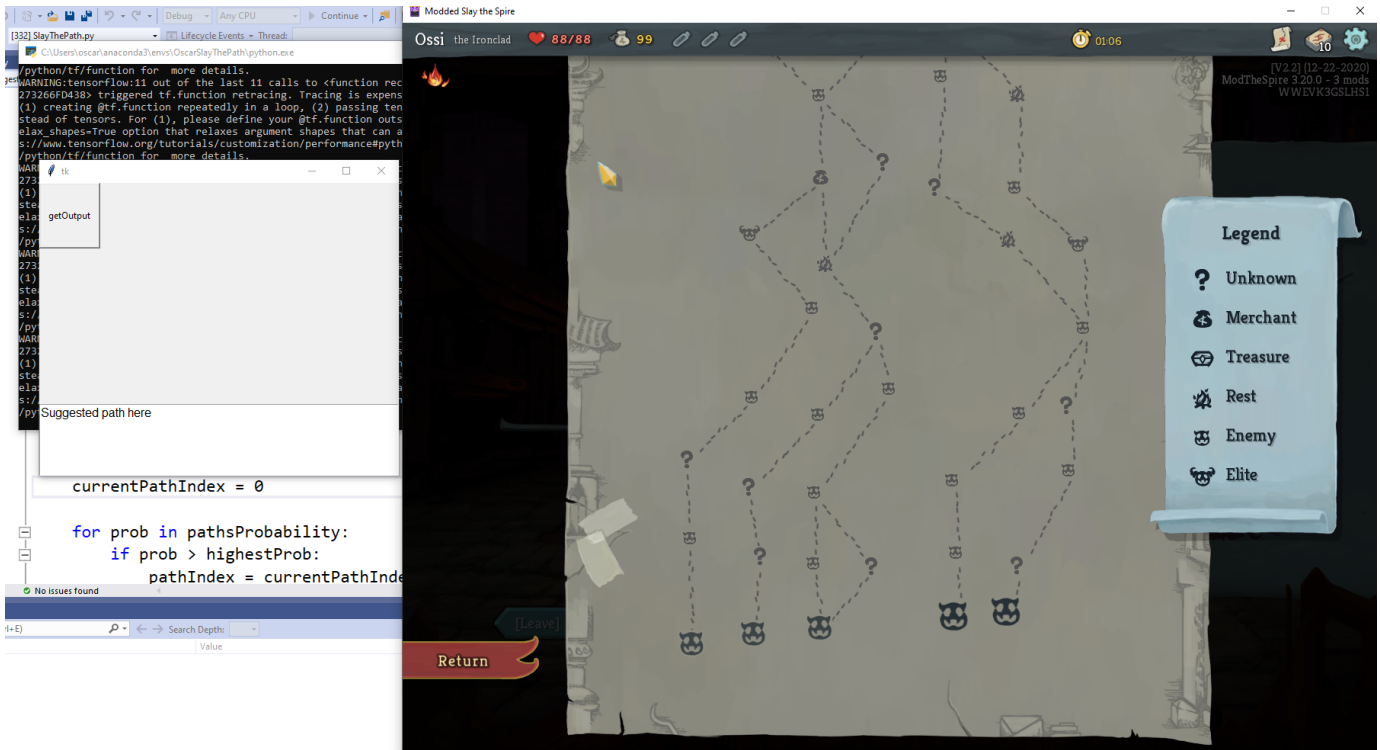


Fig. 6: This is how the setup of the programs looked. To the left the UI of the network is displayed with the button (getOutput) that was pressed after every floor to predict a new path and the text box where the path in text form was displayed. To the right is the game of Slay the Spire.

agreed that the participant was knowledgeable about how to build a deck with good synergies and how to navigate combat with those synergies.

When choosing path the participant tried to pass through as many event floors as possible (the symbol is a ?) since they felt they made them stronger than regular monster floors. They tried to take a path with at least one elite, preferably the elite was after a campfire which would mean the player could rest and heal before the hard fight. When there was two elites in a row on a path without a campfire in between the participant avoided that path since it was deemed to dangerous. The participant enjoyed choosing paths with many crossroads, that meant they could adapt their path based on how much health they had, how much gold they had and how strong they felt their deck was. If they felt healthy and if they had a strong deck they chose an aggressive path with multiple elites. If they had low health and a weaker deck they chose a safer path without elites, or at least without multiple elites.

C. Participant 2A

Participant 2A played with help and reached the 41:st floor where they died to the elite called "Reptomancer". The participant played on the seed: 1872UXKXPBRPB. The path the participant took was the following:

- Act 1: M-M-?-M-\$-R-M-E-T-?-E-M-M-M-R-BOSS
- Act 2: M-M-M-?-M-R-E-R-T-M-M-E-M-M-R-BOSS
- Act 3: M-M-M-M-?-R-E

The participant had only played Slay the Spire for 3-4 hours prior to the experiment. They had defeated the game at least one time. They felt like a beginner at the game with okay understanding of the mechanics but not a full understanding of the synergies existing in Slay the Spire. The observer thought that they played okay, it was noticeable that they had experience from other strategy games and they could confidently navigate the combat with some understanding of synergies between cards. They had a harder time building their deck and ended up with an aggressive deck with a lot of attack cards but few defensive cards.

When choosing which path to take the participant focused on the upcoming five floors and choose a path based on those five floors and not the whole path to the boss. They preferred to visit the store floor late in the act when they had amassed a large sum of gold.

The participant thought the paths provided by the network seemed reasonable and there was not that much they would have done differently if they had chosen themselves and they did not know if the different choices they would have made would have lead to better results. They noted that the path from the network prioritised elites heavily even above a path with a store on it when the participant had a lot of gold, which they felt was wrong.

D. Participant 2B

Participant 2B played without help from the AI and died at floor 33 from the boss of the act, Bronze automaton. The participant played on the seed: 1872UXKXPBRPB. The path the participant took was the following:

Act 1: M-M-?-?-M-R-M-M-T-?-?-E-M-?-R-BOSS

Act 2: M-\$-M-?-?-M-R-?-T-M-\$-M-?-R-BOSS

The participant had played for 12.6h, had reached ascension 2 and where fairly experienced not making many mistakes. The participant said they prioritised elite first but also ? rooms. If the player didn't have a lot of hp then the choose the ? rooms instead of elite. The participant ended up choosing a lot of ? rooms due to being continuously on low hp. Their reasoning for why they choose ? rooms was that they usually contains upgrades for the player without having to fight for them.

E. Participant 3A

The participant used the AI for picking a path and reached floor 31 where they died from a centurion and mystic. The participant played on the seed: 4VTLG567PJ0C3. The player reached the second act with the following path:

Act 1: M-M-M-M-\$-R-?-R-T-M-E-M-E-R-BOSS

Act 2: M-M-M-M-?-E-?-R-T-R-M-E-M-M

The participant had played the game for approximately 45h and reached ascension 1. They didn't considered themselves comfortable with all mechanics and admitted it had been a while since they last played the game. They seemed unsure on what card to pick when given a choice of cards which resulted in none of the cards providing any synergy with each other. The player also did some mistakes during combat that cost them a lot of hp. While playing the game, the participant commented that their play style wasn't as aggressive as the AIs and would if given the choice avoid elite room a lot more than the AI. In the questionnaire afterwards the participant thoughts echoed what they had said during the game, they preferred a more safer play style in which the only went into elite rooms when they had high hp. The participant did not agree with the AI choices and they thought they were too reckless. Their own preferred strategy was to keep their hp as high as possible all the time. This map contained very little opportunities for acquiring artifacts to become stronger.

F. Participant 3B

The participant played without help and managed to reach the 23:rd floor where they died to the elite called "Book of stabbing". The participant played on the seed: 4VTLG567PJ0C3. The participant took the following path:

Act 1: M-?-M-M-\$-R-?-M-T-R-M-?-R-E-R-BOSS

Act 2: M-\$-M-?-?-E

The participant had played Slay the Spire for 66 hours prior to the experiment and they had reached ascension 3. They considered themselves average at the game and thought they understood the synergies in the game although it had been a while since they played with regularity. The observer thought the participant played okay and showed a good understanding of the game.

When choosing a path through an act the participant wanted at least one elite floor but avoided two elite floors close together since that was deemed to hard. The participant localized areas of the act where there were many interesting floors with many different pathing options through those floors. This meant that the participant could adapt their path through this interesting cluster of floors based on their health, gold, deck, relics and potions, The participant made decisions to path to elites if they felt strong enough to beat the elite, which they based on their current health, how strong their deck was and if they had any good potions to use. Paths with a shop were only chosen if the participant had a lot of gold otherwise they were avoided. The participant preferred event floors over monster floors and chose path based on that when possible.

The participant had a hard time putting together a strong deck and had no powerful synergies between their cards and only four relics when they died in act 2.

G. Tables

Table I displays the result of the user study. In it we see that the players that played with help from the network outperformed the other players in 2 out of 3 pairs, in the last pair the players tied. Tables IV, V and VI shows the paths the players took through the three different acts. Table II shows the paths the ANN predicted in the simulation experiment. The column 'Most elites' shows if the chosen path was the one with the most elites taken of all available paths in the act or if there were other paths with more elites available. We see that in 8 out of 10 of the chosen paths the network has chosen a path that is tied for most possible elites in that act.

Participant	Seed	Used AI	Floor
1A	XWZX4NYBCMC8	Y	50
1B	XWZX4NYBCMC8	N	50
2A	1872UXKXPBRPB	Y	41
2B	1872UXKXPBRPB	N	33
3A	4VTLG567PJ0C3	Y	31
3B	4VTLG567PJ0C3	N	23

TABLE I: The participants number, the seed they used, if they used the AI and how far they reached

Nmbr	Chosen path	Most elites	Seed
1	M-?-M-M-M-R-E-R-T-R-?-E-?-M-R-BOSS	Tied	F35KQ198YXY8
2	M-?-?-M-M-R-E-R-T-E-?-M- <i>-M - R - BOSS</i>	Tied	2F6TQWX1FTUVP
3	M-M-M-M-?-E- <i>-E - T - R - E - M - E - M - R - BOSS</i>	Tied	25C3ETUAEW9S0
4	M-M-?-M-?-E-R-E-T-M-M-R-M-M-R-BOSS	Tied	XAGZ2Q2FVFRN
5	M-M-?-M- <i>-R - E - R - T-?-? - R-? - M - R - BOSS</i>	No	SM75VJTGRFB4
6	M-?-?-M-M-E-R-M-T-M-?-E-R-M-R-BOSS	Tied	4PDSPTBWDNPD2
7	M-M-M-?-M-E- <i>-E - T - M - E-? - M - M - R - BOSS</i>	Tied	2ME8MYG17LB6Q
8	M-M-?-M-?-E-R-?-T-M-E-R-M-E-R-BOSS	Tied	1KX3JC1SIGI5H
9	M-M-M-?-?-E-R- <i>-T - R - M - M - E - M - R - BOSS</i>	Tied	4QISQ5SD28TRB
10	M-M-M-M-M-E-M-E-T-R-M-R-M-M-R-BOSS	No	4B2FPXUWQKGG4

TABLE II: The results from experiment trying to find if elite rooms is a common factor when the AI decides the best path

Average number of room type in Act number	Winning runs	Losing runs
Average number of Elites in Act 1	1,8471612	1,319591
Average number of Elites in Act 2	1,6750392	0,9974074
Average number of Elites in Act 3	1,5928123	1,0254569
Average number of Campfires in Act 1	2,8800755	2,072388
Average number of Campfires in Act 2	2,934879	1,8063782
Average number of Campfires in Act 3	2,8433645	1,8485634
Average number of Shops in Act 1	0,9148667	0,7350115
Average number of Shops in Act 2	1,1808949	0,8789928
Average number of Shops in Act 3	1,2618134	0,9138464

TABLE III: Table that shows the average occurrence of certain room types in different acts

H. Average occurrence of specific room types

Data analysis on the data provided by the developers of Slay the Spire was performed with the intentions of analysing the importance of certain room types on winning paths. A total of 1008636 runs chosen at random from the data set was analysed, where 93242 runs were wins and 915394 were losses, this amounted to a win-rate of 10,186%. Table III showed the average occurrence of the room types elite, campfire and shop through acts 1,2 and 3. In the table we saw that in winning runs the average number of elites taken was higher than in losses through out all acts. The average number of campfires was significantly higher, approximately one entire campfire more in winning runs. The amount of shops visited was close but winning runs still visited more on average. The average amount of shops was close to one per act for winning runs. Something to keep in mind was that in losing runs the player could have died before the boss and then would not have passed through the same number of rooms as a player in a winning runs.

IX. DISCUSSION

A. Network

The tests performed with the network showed that the players using the ANN for path finding reached a slightly higher floor than the players not using the ANN. The constructed Network works but results from testing suggest there are more ways to improve it to get the best path possible. Ways to improve it could be by dividing complex data that are difficult for the network to understand into smaller units that are easier to process or adding more input to make the evaluation better.

The data gathered show that everyone except one person from the group using the AI reached a higher floor than the control group without the help of an AI (see I). The person that did not reach further tied with the person not using the AI that played on the same seed. They both reached the last boss but the person that played with the AI defeated the boss while the other person did not.

The results from the test are promising and shows we might have succeeded in developing an AI that can help players picking a path. Unfortunately due to the pandemic we didn't manage to gather a very large group of people to test with which meant there isn't enough data to plot a definitive connection to if the ANN really helped.

While the network we constructed worked it still behaved problematic in a few ways. For example, when the network had found a path for the player then it never changed to anything else for the whole act. Even if the player had 5 hp left and had to choose with either following the old path going towards lots of elites or taking a new easier path with a lots of opportunities for healing it still kept to the old path it had chosen in the beginning of the act. This is something we found strange since the hp was one of the inputs the ANN had been training with.

Could there be something wrong with the network or had it found pattern we humans hadn't thought of, a strategy for the AI? Since the result the AI was trying to reach was if the

person won the whole game or not then maybe it had a more long distance goal in mind than just surviving the current fight. To win the whole game you need a lot of good items like relics which are dropped from '?' rooms and elite monster. Maybe the AI prioritised trying to win the whole game by attacking as many enemies as possible to gain as many strong items to increase the chances of winning the whole game?

The AI also seems to not have put a lot of weight to when players started to have a lot of gold. There were times when the players had a large amount of gold with a shop room next to them but the AI told them to choose a different room. This seems to be the same problem as when the ANN kept to the same path even when low on hp, the ANN does not adapt to the situation. One thing that could have had an impact on the poor adaptability was that the ANN was trained on 24 input nodes where 15 of them were the rooms on the path but only one was for gold and one was for hp. This might have meant that the inputs for gold and hp did not have enough impact on the final output, this would explain the poor adaptability. Furthermore the inputs for gold and hp were integer values that were normalized and the difference between 20 gold and 200 gold might not have been enough to impact the output even though the difference in the game is big. To combat this another version of inputs could be to divide hp and gold into three columns each where the columns are: low gold/hp, medium gold/hp and high gold/hp. Then the thresholds for low, medium and high could be decided based on experience. If the player has for example 300 gold which is a lot, the input for "high gold" would be 1 and the inputs for "low gold" and "medium gold" would be 0. This could have more impact on the final output and lead to better adaptability.

To alleviate the problems found while testing there were a few ideas that came to mind while evaluating the results. We had done a few smaller test of the program with different numbers of neurons and layers but we still had many more complex types and shapes we could have explored.

Was the loss function we used the right one?, this is another venue that could have been explored by trying other functions such as categorical crossentropy which would have been useful if 2 or more labels were used. Evaluating more labels at the same time meant we could have sent in all possible paths at the same time to the ANN, and this would led to shorter time computing an answer from the artifact. The AI could furthermore have been tested with different variations to each layer to find a more fitting solution with better outputs.

Another large factor we think might could have done to make the AI more accurate is dividing up the inputs for the deck and relics differently. The inputs we used now was to cram the deck into a whole string and the relics into another string and then send them off to the AI. There might be a problem when the preprocessing layer transformed the string input into a number. How the string is transformed to a number is not something we knew a lot about and it might been too difficult for the AI to evaluate such a complex string of characters from just one number since the string variations can get so varied due to the large number of cards a deck

can contain in infinite variations. The number the string is transformed into might not be a fair representation of the data.

This problem could have been solved by dividing up the deck input into smaller inputs for each card since this is something we did for the path which resulted in a dramatic improvement of decision making of the ANN. The problem with this solution is that the input for the deck will be of a varying amount and no one from the team had the knowledge this late into the project to know how to handle varying inputs in machine learning.

A second idea for improving input from the deck is an idea for when it was trained. The deck input the ANN is trained with contains all the cards the player used at the end of the run. Because of this, the deck input is not very useful in the middle of a floor since the deck data it has been trained with is from the end of the run. To improve the training the ANN could be provided with how the deck looked at the floor it's currently on and not how it looked at the end of the whole run. The reason we did not do this from the start is that this is much harder to gather from the data provided by the developers and we did not have the time, since we had to have the artifact ready for the experiments in time.

The ANN was trained on data from all classes and an idea for improvement would be to only train the ANN on data from Ironclad since that is the class used in the experiment. There might have been some strategies that it learnt from data of runs of other classes and that is not optimal. However the current ANN has the advantage that it is general and can be used for all classes which is also important.

The ANN's evaluation goal was to reach number that was between 0 and 1. 1 meant the recorded run had won the whole game, 0 meant it was lost along the way. This might not be the best way to evaluate a successful run. Another better way to evaluate the best path might have been to look at how many floors the player managed to reach in each run. The amount of floors reached is a more gradually increasing number that could have made it easier for the ANN to classify how successful each path is.

There were still a lot of different variables from the games we didn't choose to add. Examples of input that could have been added is potions which can have a drastic effect on the possibilities of winning. Potions doesn't help in every match but is of great help when used in key fights with powerful opponents like elites and bosses where you might just need a small push to get through them. Another example of a missed input is the bonus the player gets when starting a new run which won't have a large impact on the path if you choose to start with extra hp but could make a big impact if the player chooses the bonus that makes the next 3 rooms of enemies start at 1 hp. Since the bonus also applies to elite the benefits are obvious and would basically guarantee the player a free artifact.

B. Simulation experiment

The smaller experiment that simulated ten predictions for ten different playthroughs of the game showed some interest-

ing patterns in how the ANN chooses paths. Our hypothesis before running the experiment was that the ANN would choose the path with the most elite fights. In this subsection we aim to discuss the patterns we have found and if our hypothesis was correct.

A pattern that was noticeable was that all of the ten paths had at least one elite on them and even further, nine out of the ten paths had multiple elites, with the majority of them having two elites (six out of ten paths), the table II shows all the predicted paths in the experiment. When we analysed the data provided by the developers of Slay the Spire it showed that on average a winning run took 1.85 (see table III) elites in the first act which was slightly less than how many elites the ANN wanted to take. The data also showed that on average winning runs took more elites than losing runs, however something to keep in mind is that losing runs often end before all rooms in the act are visited. What we gathered from this was that elite fights and more specifically the reward of a relic after the elite fight was very important to the success of the playthrough. Something we took into account was that the ANN was only interested in the binary outcomes winning or losing and did not care how far into the playthrough the player made it. This might have meant that the ANN predicted harder paths with more elites because if the player made it through them they would have been a lot stronger and have a higher chance of beating the game. If instead they died on the hard path it would have been the same as dying in a later act because the player was not strong enough. A very high risk and high reward play style seemed to be what the ANN wanted.

The ANN seemed to prefer paths with multiple campfires since in 8/10 cases the predicted path had 3 or more campfires. This was similar to what the data analysis of the data provided by the developers showed. It showed that the average number of campfires taken was between 2.84 and 2.93 (see table III) for winning runs. Something to keep in mind was that every path had a campfire at the end right before the boss fight of the act. Campfires were very useful either for the heal the player could have chosen if they were low on health or the upgrade to a card they could have chosen, which was a very powerful effect that certainly made the deck stronger. Since campfires were so useful it made sense that the paths that were predicted had multiple of them.

Something that was not prioritized was shops. In only half of the predicted paths a shop was passed through and there were never multiple shops on a path. The paths were predicted at the start of the first act when the player did not have a lot of gold and the ANN never changed its predictions during the playthroughs. Therefore it was not strange that the ANN did not prioritize shops when it predicted paths for this experiment. However the data analysis in table III showed slightly more prioritization on shops with 0.91 on average for the first act. The data analysis and the simulation experiment both showed that multiple shops per act, especially in the first act, was not an optimal strategy. Something to keep in mind was that the ANN did not take into consideration future gold earnings from the floors on the predicted path, which if it did might

have meant that it would have changed its path to a path with a shop on it in more instances to be more similar to what the data analysis showed. There might have been something deeper that the ANN found in that shops are not that important when paths were chosen and that a higher priority should have been on paths with elites and campfires. There was also always a possibility of getting a shop from the regular question mark events which might have had something to do with having lower priority on shops.

The hypothesis we had going into this experiment was: The path chosen by the ANN is the one with the most elite fights. It was not entirely correct since only 8/10 of the predicted paths tied for being the path with the most elites in comparison with the other choosable paths in that specific case. The other two predicted paths were not the paths with the most elites for those specific cases. What was correct was that elites play an important role when choosing paths and as stated before nine out of ten paths had multiple elites on them. Elites were important to the ANN but not the only thing it took into account since it did not always predict the path with the most elites.

C. Participants thoughts about paths

The participants of the experiment had both similarities and differences in how they viewed paths compared to the ANN. The participants all thought elites were important and tried to choose at least one when deciding paths, the participants paths can be seen in the tables IV, V and VI located in Appendix D. They were however more careful than the ANN and rarely chose multiple elites in an act which was also against what the data analysis showed (see table III). The participants were also more careful in how they adapted their paths towards more or fewer elites depending on how much health they had and how strong they felt. Something two out of the three that used the ANN thought was that the ANN was too risky and chose paths with too many elite fights. This high risk and high reward playstyle was noted in the other simulation experiment as well. The result of the experiment showed that in all three cases the participant with the help from the ANN reached further and the only win came from a player that had help from the ANN. This supports the idea that the high risk playstyle of the ANN was better than a more careful approach which the participants preferred. For a definite conclusion more testing would be required.

A big difference between how the participants thought about paths and how the ANN thought about paths was in how much priority the participants had on the ability to adapt their paths throughout the act. All the participants stated that they adapted their path throughout the playthrough based on variables but the ANN did the opposite and stuck to the same path regardless. This might have been a fault with the ANN but it could also have been that the ANN thought that sticking to a single path the entire time lead to more success. Sticking to the same path could have meant that the player instead adapted their choices in what cards they picked up and what event choices they made after how the path looked. The participants

instead preferred adapting their path based on their health, gold and battle strength. Furthermore multiple participants made pathing choices based on the upcoming couple of rooms to usually the next crossroad and then made new decisions for the next upcoming rooms, they basically divided the entire path from start to boss into smaller blocks. Based on their health, gold and battle strength they chose different blocks. This was not something the ANN did but it could have been another interesting way of building it.

Three of the six participants said unprompted that they preferred paths with many question mark events over the paths with regular monster events. The ANN did not seem to have any priority on question mark events over monster events or vice versa instead it prioritized elites and campfires heavily and it did not seem to care if the other parts of the path were monsters or question mark events.

D. contribution to other fields

Analysing the results from the experiment will give benefits to many different actors like the players, game designers and researchers. For the players the results may offer more insight into what decisions the players should make to perform better. By learning what rooms are most usable in different situations the player will become better at playing the game.

The researchers can use the experiments to get new insights in how to study patterns in decisions making systems. The results from the experiment can be used in other research too, for example a study could be done to look at how a specific kin of room like the the enemy rooms affect decisions making process.

Furthermore, the results from the experiments can also be used by game designers. Analysing the data gathered for experiment 2 can prove useful to game designers for learning how to use paths in their own game maps to create a more fun game experience. The results from the experiment with the network also offer insights the designers can learn from to use for paths. The network has prioritised rooms differently which can be further studied by the designers to improve their own knowledge about paths in games. The data can be used to show what rooms are important so they learn from this when balancing their game. Maybe even the AI itself could be used for when they need to quickly test the map component of their game to reach further conclusions about how the player will play their game? The network could be used as a test tool to quickly find strategies the players and analyse them to further improve the game designers own games. One thing to note is that the AI as it probably wont be smart enough to find all ways to play but can instead help cover the larger more general parts of play testing to improve the effectiveness of play testing.

E. Summary of findings

This section summarizes the answers to our research question and how this study is a contribution to Slay the Spire.

RQ1: How can machine learning help predict successful paths in Slay the Spire?

The artifact that we developed has shown to be helpful with predicting paths and performed better than human decision-making when we did our experiment. More testing is required to state with certainty that our artifact is better at choosing paths than humans but what we can prove is that the artifact does provide paths that are useful and can help the player win the game. The priority the ANN shows towards elites and campfires is similar to what the data analysis shows for winning paths which means the ANN does something right in its decision making. Our machine learning approach is quite basic and yet we could make an artifact that is helpful in predicting paths. With more fine tuning and more complex versions of the artifact it should be possible to make an even better artifact more helpful to the player. With regards to the research question we believe we have proven that machine learning can be helpful to the player when they are deciding on what path to take.

RQ2: What do successful paths have in common with each other?

Through the simulation experiment and the interviews with the participants we gathered data about how our artifact thought successful paths should look and how the participants thought successful paths should look. The artifact prioritized elite floors heavily and tried to pass through multiple per act. It had a very high risk and high reward approach when it predicted what path to take. The participants also liked passing through elite floors but they were more cautious than the artifact and preferred only fighting elites when they felt healthy and strong enough. Despite the differences in approach we can say that elites are important to a successful path for both the artifact and the participants, exactly how many the player should aim for is harder to say definitely. This is further corroborated by the data analysis that shows that on average the number of elites for winning runs is in between how many the ANN chooses and how many the participants chooses (see table III). The artifact performed better in our small test and this might mean that its high risk approach is better but there would need to be more testing. Slay the Spire is also a very adaptable game which was primarily echoed by our participants and it may be that the answer to how many elites is enough, is answered differently depending on a few variables such as health and strength of deck.

Both the artifact and the participants prioritized campfires which we could see in the simulation experiment in table II and in what paths the participants chose which can be seen in the tables in Appendix D. Furthermore the data analysis shows that campfires are important and on average 2,9 are taken per act in winning runs. Based on all of them both prioritizing campfires we can say that campfires are important to a successful path and they should be taken into account when choosing what path to take.

There were conflicting thoughts regarding shops on the paths. The artifact did not prioritize shops at all and it was more of an afterthought if they should be on the path or not. It also did not alter its path towards a shop when the player had a lot of gold which may be a sign of it not thinking that

shops are as important as for example elites. The participants thought shops were useful especially late in the acts when the player has a lot of gold to spend and they would adapt their path towards a shop if they had a lot of gold. The data analysis shows that on average one shop per act is taken in winning runs which is more similar to how the participants prioritizes shops. The conclusions we can draw regarding shops is that one shop per act seems to be optimal based on the participants opinions and the data analysis but more importance should be shown towards choosing a path with elites and campfires since all of them prioritize those. It may also be something that the player can adapt towards late in the act if they have a lot of gold and not something they plan from the start.

To answer the research question we can state with certainty based on our experiments that a path should have at least one elite to be successful, more elites than one depends on health and other variables. Or we need more testing to prove that the artifact's approach of choosing multiple elites is the correct approach. We can also state that campfires are important to a path and a successful path should try to gather multiple which is corroborated by the ANN:s predictions, the opinions of the participants and the data analysis.

This is a contribution to how paths and pathing decisions are viewed in Slay the Spire and it may help players understand what is important for a path in Slay the Spire. This analysis may also be important for future studies about pathing in Slay the Spire.

X. CONCLUSION

This paper set out to show that machine learning can help with predicting paths in Slay the Spire through an Artificial Neural Network that predicts the win rate of specified paths, we have found that the Artificial Neural Network we made can help with predicting paths in Slay the Spire. While we have specifically focused on the success of predicting paths with a machine learning approach, the findings we have made regarding patterns in successful paths are likely to be of importance to other research about paths in Slay the Spire.

A. Future work

The research done has shown good results but more work is possible in the future. More extensive testing could be done to test larger groups of players. Since the Covid-19 pandemic limited communication with people the test group was smaller and more difficult to gather than what it usually would have been. There could be more time and work added to analyse what strategies the AI uses to understand what is required for a good path.

As discussed in the discussion, more work could be done on the input to the ANN. Since the input for relics and cards seemed flawed and didn't contribute to the network the right way, a way to handle the varying inputs for these variables could improve the network.

Another way to improve the network itself is looking to continue develop the network itself to a more complex type for example by seeing what adding more hidden layers do or

adding input that can be dynamically changed to fit the number of different cards a deck contains. As it is now, the network is very rudimentary and we have no doubts if more work was done in this area then large improvements could be made.

But while the network we made works and gives us result this isn't the only way to use a computer to help pick a path. ANN is just one way to pick a path for the players and many more ways could be used like for example a decision tree. To truly know if machine learning is the best way to create a program for picking paths, all other ways has to be tested too. Using a decision tree formed from gathering data about strategies from interviews with very experienced players (ascension level 15+) could also be a very effective way to find the right path and should be done if the effectiveness of the ANN is to be truly evaluated as a effective tool.

An interesting version of the ANN would be to only train it with successful runs. Now we train it with both successful runs and failed runs to have it learn as broadly as possible. But if it is only trained on successful runs the ANN should in theory learn more correct behaviour than the current ANN, since it will only learn from successful strategies. The current ANN however learns what choices lead to failure as well as success which could also be important. It would be interesting in the future to compare two versions of the ANN where one is trained on only successful runs and the other is the current version.

Lastly another interesting experiment that could been done is training multiple networks with different batches of data to analyze their different behaviors. Looking at the different behaviors between an AI trained with the data from only inexperienced players of ascension level 0 and compare with an AI trained with experienced players of ascension level 20 could be very interesting. The different strategies the AI's would develop might be very different or the same. Maybe there is a drastic difference between the different players strategies? Or maybe there is just small differences and how successful the players are is mostly how good they are at picking the right cards and how they fight?

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APPENDIX A
PARTICIPANT QUESTIONNAIRE BEFORE THE EXPERIMENT

2021-05-17

Slay the Spire: Experiment deltagare

Slay the Spire: Experiment deltagare

Discord namn som man kan nå dig på

Ditt svar

Hur mycket har du spelat Slay the Spire tidigare? (Gå gärna in i spelet/steam och kolla)

Ditt svar

Hur erfaren känner du dig som Slay the Spire spelare?

1 2 3 4 5 6 7 8 9 10

Helt ny med spelet. Har bara spelat ett par gånger och aldrig besegrat spelet.



Expert som kan allt om spelet.

Känner du dig bekväm med mekanikerna i spelet? (Vad de olika karaktärerna har för skillnader. Vad olika kort/reliker gör och vad de har för synergier mellan sig)

Ditt svar

Om du har besegrat spelet. Vilken ascension level har du nått som högst?

Ditt svar



APPENDIX B
QUESTIONNAIRE FOR AFTER THE EXPERIMENT

2021-05-21

SlayTheSpireFrågeställningar

SlayTheSpireFrågeställningar

1. Tog Ain en annorlunda väg än vad du skulle gjort själv? På vilket sätt? Beskriv

2. Var det något val i vägen som du inte håller med om?

3. Var det något val i vägen du tycker gjordes bra?

4. Hur resonerar du när du väljer en väg? Tex försöker du möta så många elite som möjligt.

Det här innehållet har varken skapats eller godkänts av Google.

Google Formulär

APPENDIX C

PROTOCOL FOR AN EXPERIMENT SESSION

How do we test?

- Start the ANN program, the mod and Zoom
- Give the participant remote access
- The observer writes down the following: How far the participant made it, how they lost and any interesting things that happened.
- Do not comment on what the participant does, only observe.

Questions to ask during the playthrough:

- Did the AI take a different path than what you would have taken? What was different?
- Some choice the AI did that you were against?
- Some choice you think the AI did well?
- What do you think about when choosing a path?
 - Do you try to meet a specified number of elites?
 - What is your overall pathing strategy?
 - Ask follow up questions to get a discussion on how the participant chooses a path.

APPENDIX D
PATHS THE PARTICIPANTS TOOK

Participant	Path act 1
1A	M-M-M-M-M-R-E-R-T-R-E-R-E-M-R-BOSS
1B	M-?-?-?-R-?-E-T-?-M-M-?-?-R-BOSS
2A	M-M-?-M-- $R - M - E - T - ? - E - M - M - M - R - BOSS$
2B	M-M-?-?-M-R-M-M-T-?-?-E-M-?-R-BOSS
3A	M-M-M-M-- $R - ? - R - T - M - E - M - E - R - BOSS$
3B	M-?-M-M-- $R - ? - M - T - R - M - ? - R - E - R - BOSS$

TABLE IV: The participants path though the first act

Participant	Path act 2
1A	M-?-M-M-?-R-M-?-T-?-E-- $M - E - R - BOSS$
1B	M-M--?-?- $R - ? - R - T - M - E - M - ? - E - R - BOSS$
2A	M-M-M-?-M-R-E-R-T-M-M-E-M-M-R-BOSS
2B	M--M-?-?- $M - R - ? - T - M - M - ? - R - BOSS$
3A	M-M-M-M-?-E-?- $T - R - M - E - M - M$
3B	M--M-?-?- E

TABLE V: Second acts path

Participant	Path act 3
1A	M-M-?-M-M-E-M-E-T-E-?-R-M-M-R-BOSS
1B	M-?- $M - ? - R - M - E - T - E - ? - ? - R - M - R - BOSS$
2A	M-M-M-M-?-R-E
2B	-
3A	-
3B	-

TABLE VI: Third acts path